

How Accurate Are Non-Invasive BLOOD GLUCOSE METERS



Smita Dikondkar is pursuing PhD in biomedical electronics from University of The South Pacific, Fiji

Non-invasive blood glucose testing makes traditional blood sampling a thing of the past. This technology allows monitoring of blood glucose levels without piercing the skin

There is a precise difference between glucose concentrations (mg/dl) when blood sample is taken from arteries, capillaries and veins of the same subject due to biologically dynamic conditions. Diabetes healthcare solutions must be accurate and precise for glucose meters, as an insulin dose is decided for diabetes mellitus patients undergoing insulin therapy, which is thoroughly based on clinically-accepted blood glucose (BG) meter ranges. It is important that the measured BG falls within the clinically-acceptable accuracy zone and, thus, acceptable error grid.

It is important to oversee the evaluation of calibration and accuracy of non-invasive BG meters that are used for BG concentration measurements and the current state of art.

Trueness is defined as the closeness of agreement between the average value obtained from a large data of repetitive tests and an accepted reference value according to ISO standard 5725. It is also expressed as bias. Systematic error of a method is usually estimated by determining its trueness.

Clarke Error Grid Analysis (EGA) was developed in 1987 to evaluate the clinical accuracy of a patient's BG estimates compared to the BG values obtained using a glucose meter. It was then used to quantify the clinical accuracy of the BG estimates generated by meters compared to a reference value. Description of EGA appeared in Diabetes Care in 1987. Eventually, EGA was accepted as one of the gold standards for determining the accuracy of BG meters.

Fig. 1 shows Clarke EGA scatter plot of a reference glucose meter and an evaluated glucose meter, divided into five regions, namely, A, B, C, D and E. The scatter plot under region A shows values within 20 per cent of the reference glucose sensor, and B contains points outside 20 per cent. However, these are accepted as valid for appro-

priate treatment. Points under region C lead to inappropriate treatment, and points in region D are potentially erroneous to determine hypoglycemia or hyperglycemia. Region E includes the points that would mislead the treatment for hypoglycemia or hyperglycemia.

ISO standard 15197:2003 for BG monitoring true value performance is based on a consensus of professionals around the world. It is a clinically-recognised standard in many countries, along with being Food and Drug Administration (FDA)-recognised consensus standard in the USA. While evaluating BG accuracy, subjects having different diabetes types, ages (18 to 85 years), sex and ethnicity were selected. Fingertips and forearms were taken as measurement sites.

There are two prominent methods used in BG measurement: colour reflectance method (photometric) and amperometric method (electrochemical sensor technology). Analogue front-end reflectance method uses optical sensors and amplifiers to determine BG.

Invasive blood glucose meters

Invasive glucose measuring meters follow amperometric or photometric end-point working principle. A blood sample of 2µl - 3µl of the subject is directly used in the procedure. Depending on the type of diabetes mellitus of the subject, BG is measured at least one to four times a day.

The analogue front-end of the reflectance method uses optical sensors such as LEDs, photo transistors and a trans-impedance amplifier. It uses the colour reflectance method based on reaction colour intensity in the reaction layer of the test strip by reflectance photometry. The meter quantifies the colour change and determines a numerical value that represents the concentration of glucose (mg/dl) in blood.

Non-invasive blood glucose meters

Modern technology-based glucose meters are innovative. These are also known as

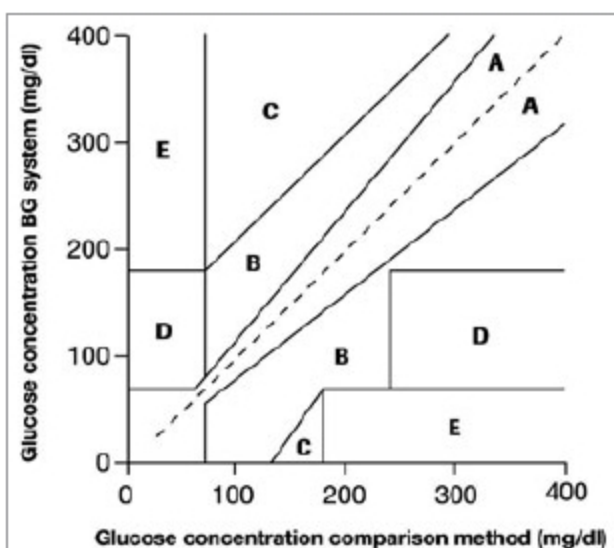


Fig. 1: Clarke Error Grid Analysis scatter plot